

Experiment: 9: Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1

```
clc;
clear;
close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

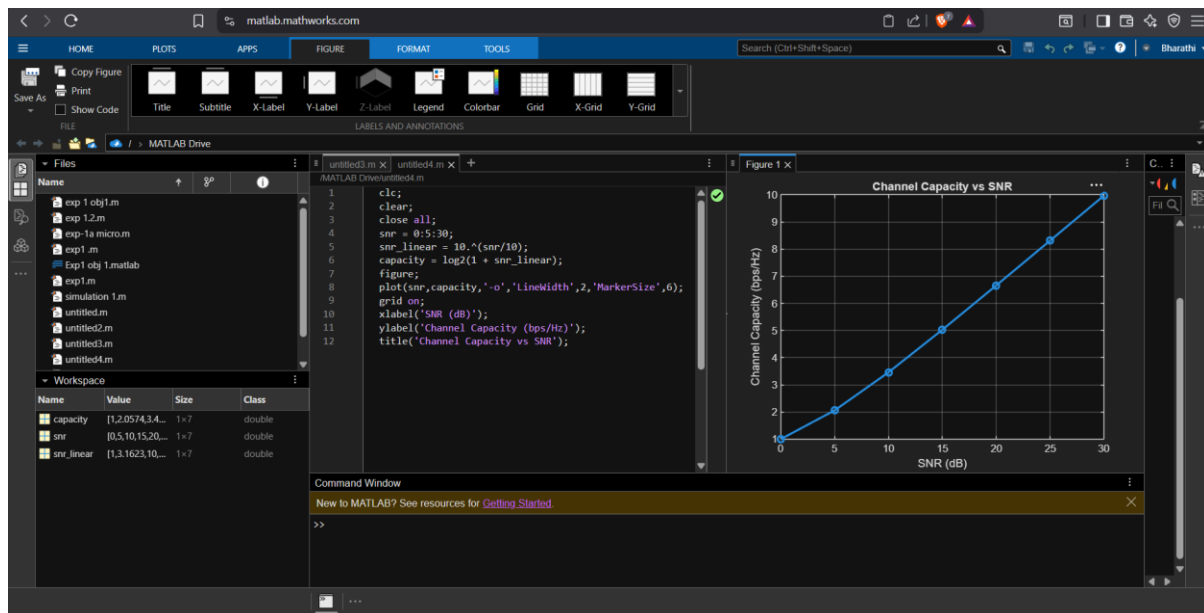
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```



Objective – 2

```
clc;

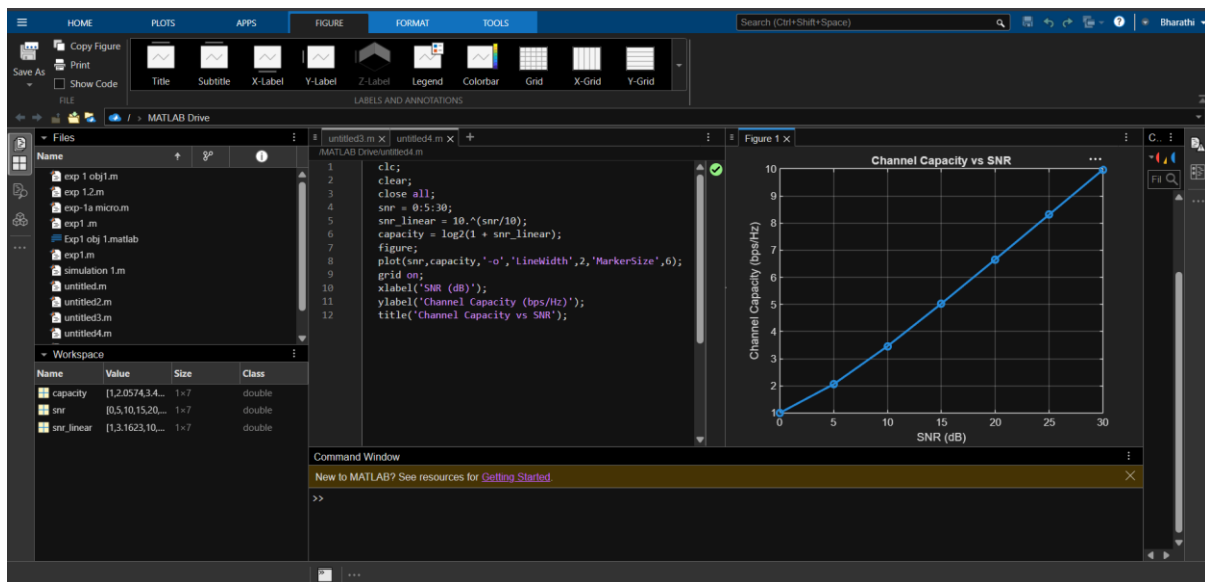
clear;

close all;
```

```

snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');

```



Objective – 3

```

clc;
clear;
close all;

antennas = [4 8 16 32 64]; % Number of Antenna Elements gain =
10*log10(antennas); % Antenna Gain (dB)

figure
bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')

```

title('Antenna Gain for Different Antenna Array Sizes')

